

Espacios de sucesiones

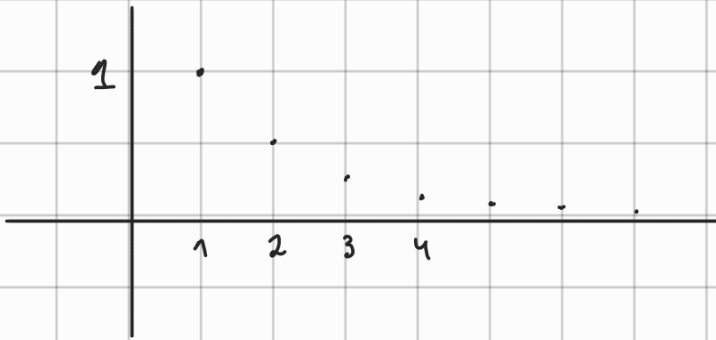
NOT: $(a_n)_{n \in \mathbb{N}} := a \in l^\infty$

acotadas

Def: Sea $l^\infty = \{ (a_n)_{n \in \mathbb{N}} \subseteq \mathbb{R} / \sup_{n \in \mathbb{N}} |a_n| < \infty \}$

definimos la métrica $d_\infty(a, b) = \sup_{n \in \mathbb{N}} |a_n - b_n|$

Ejemplos: $a_n = \frac{1}{n}$, $0_n = 0 \Rightarrow d(a, 0) = \sup_{n \in \mathbb{N}} |\frac{1}{n} - 0| = 1$



$b_n = -\frac{1}{n}$. Calculemos $d(a, b)$

$$d(a, b) = \sup_{n \in \mathbb{N}} \left| \frac{1}{n} - \left(-\frac{1}{n}\right) \right| = 2$$

Prop: (l^∞, d_∞) es e.m

Ej: Analizamos $B(0, 3) = \{ a \in l^\infty / d(a, 0) < 3 \}$ ($a, b \in$)

$c_n = 3 - \frac{1}{n} \in B(0, 3)$?

No, pues $d(0, c) = \sup_{n \in \mathbb{N}} |3 - \frac{1}{n}| = 3$

Para ver que es 3 basta ver que $\forall \varepsilon > 0, \exists n \in \mathbb{N} / 3 - \frac{1}{n} > 3 - \varepsilon$
(cosas de guía 1)

$$\Rightarrow d(0, c) = 3 \Rightarrow c \notin B(0, 3)$$

$$\text{Caractericemos } B(0, 3) = \{x \in l^\infty / \exists \varepsilon > 0 : |x_n| < 3 - \varepsilon\}$$

$$\text{Dem: } \subseteq) \text{ Sea } y \in B(0, 3) \Rightarrow \sup_{n \in \mathbb{N}} |y_n| < 3$$

$$\text{Tomamos } \varepsilon = \frac{3-1}{2} \text{ pues } |y_n| \leq 1 + 3 - 3 = 3 - \underbrace{(3-1)}_{\varepsilon} = 3 - 2\varepsilon < 3 - \varepsilon$$

$$\Rightarrow |y_n| < 3 - \varepsilon$$

$$\supseteq) \text{ Sea } x \in l^\infty / \exists \varepsilon > 0, |x_n| < 3 - \varepsilon. \text{ Sea } \varepsilon > 0 /$$

Que cumpla esto

$$\text{Sabemos que } |x_n| < 3 - \varepsilon \forall n \in \mathbb{N}. \text{ Entonces } \sup_{n \in \mathbb{N}} |x_n| \leq 3 - \varepsilon < 3$$
$$\overset{||}{\downarrow} \\ d(x, 0) \rightarrow x \in B(0, 3)$$

$$\underline{\text{Ej:}} \text{ Sea } X_n = \{a \in l^\infty / |a_n| < 1\}$$

$$\text{por ej: } \left(\frac{1}{n} - \frac{1}{2}\right)_n, \left(\frac{1}{2}\right)_n, (0)_n, \left(\frac{1}{2^n}\right)_n$$

$$\underline{\text{Pdo:}} \quad x \in \bar{X} \Leftrightarrow \exists (x_n)_n \subseteq X / x_n \xrightarrow{n} x$$

Calculamos $\overline{X_1} \stackrel{?}{=} \{a \in \ell^\infty / |a_n| \leq 1\}$

$\subseteq$) Supongamos $a \in \overline{X_1} \Rightarrow \exists (x^k)_{k \in \mathbb{N}} \subset X_1 / x^k \rightarrow a$

NOT: $(x^k)_{k \in \mathbb{N}} \subset \ell^\infty$ es sucesión

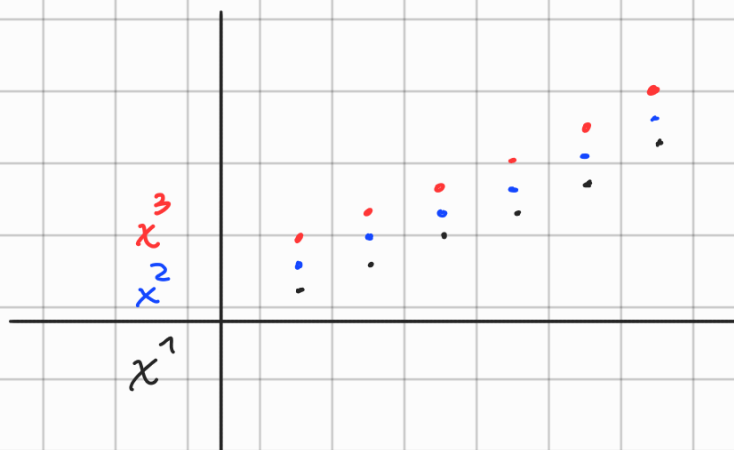
$$X_1 \ni x^1 = \overbrace{x_1^1, x_2^1, \dots}^1$$

$$X_1 \ni x^2 = x_1^2, x_2^2, \dots$$

$$\vdots x^3 = x_1^3, x_2^3, \dots$$

$$\vdots x^k = x_1^k, x_2^k, x_3^k, \dots$$

$$\vdots \overbrace{-1 < 1}^{\text{green}}$$



$\downarrow$

$$a = (a_1, a_2, a_3, \dots)$$

$$x^k \rightarrow a \text{ cuando } d_\infty(x^k, a) \xrightarrow[k]{} 0$$

$$\sup_{n \in \mathbb{N}} |x_n^k - a_n| \rightarrow 0$$

$$|x_1^k - a_1| \leq d_\infty(x^k, a)$$

$$\downarrow$$

$$0$$

$$\downarrow$$

$$0$$

Como $(x_n^k)_{k \in \mathbb{N}} \subseteq \mathbb{R}$ y $|x_n^k - a_n| \rightarrow 0 \Rightarrow x_n^k \rightarrow a_n$

$$x^k \in X_1 \Rightarrow |x_1^k| < 1 \quad \forall k \in \mathbb{N}$$

Invocamos Ej 9 de la Guía 1 que dice:

$$\left[\begin{array}{l} x_n \rightarrow x, y_n \rightarrow y, x_n \leq y_n \forall n \Rightarrow x \leq y \end{array} \right]$$

$$|a_1| = \lim_{k \rightarrow \infty} |x_1^k| \leq 1$$

$\Rightarrow$ Sea $a \in \ell^\infty / |a_1| \leq 1$, quq $\exists (x^k) \subset X_1, x^k \rightarrow a$
 $\parallel$
 $(a_1, a_2, a_3, \dots)$

Podemos definir $\forall k, \forall n \geq 2, x_n^k = a_n$

Ahora busco $(x_1^k) \subset (-1, 1)$ y $x_1^k \rightarrow a_1$

$$a_1 \in [-1, 1] = \overline{(-1, 1)} \Rightarrow \exists (x_1^k)_{k \in \mathbb{N}} \subset (-1, 1), x_1^k \rightarrow a_1$$

$\uparrow$
Pdo

$$\text{Queda ver } d_\infty(x^k, a) = \sup_{n \in \mathbb{N}} |x_n^k - a_n| =$$

$$= \sup_{n \in \mathbb{N}} \{ |x_1^k - a_1|, \underbrace{|a_2 - a_2|}_0, 0, 0, \dots \} = |x_1^k - a_1|$$

$$\text{Pero entonces } d_\infty(x^k, a) = |x_1^k - a_1| \rightarrow 0$$

$$\Rightarrow x^k \rightarrow a \text{ y } x^k \in X_1 \Rightarrow a \in \overline{X_1}$$